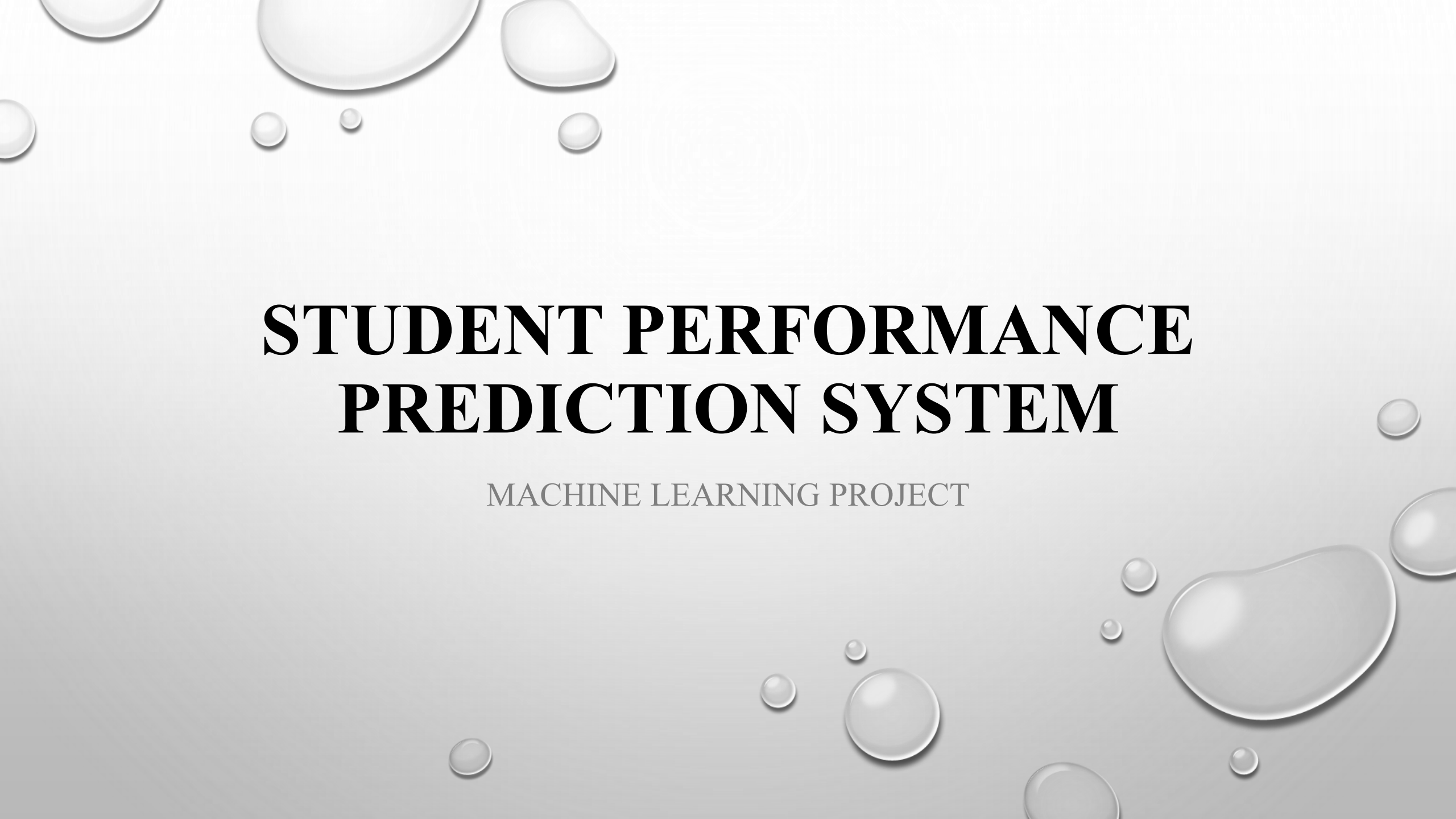


The background of the slide is a light gray gradient. It is decorated with numerous realistic water droplets of various sizes. Some droplets are large and prominent, while others are small and scattered. They are primarily located in the top-left and bottom-right corners, with a few smaller ones in the center and top-right areas. The droplets have highlights and shadows, giving them a three-dimensional appearance.

STUDENT PERFORMANCE PREDICTION SYSTEM

MACHINE LEARNING PROJECT

PROJECT OVERVIEW

PROJECT DESCRIPTION:

- THIS PROJECT PREDICTS STUDENT FINAL MARKS USING MACHINE LEARNING BASED ON STUDY HOURS, ATTENDANCE AND PREVIOUS MARKS.
- THE SYSTEM HELPS ANALYZE STUDENT PERFORMANCE AND PROVIDES PREDICTIONS USING A TRAINED ML MODEL.

PROBLEM STATEMENT

PROBLEM:

- EDUCATIONAL INSTITUTIONS NEED A SYSTEM TO ANALYZE STUDENT PERFORMANCE AND IDENTIFY FACTORS AFFECTING ACADEMIC RESULTS.

SOLUTION:

- A MACHINE LEARNING MODEL THAT PREDICTS FINAL MARKS USING STUDENT-RELATED FEATURES.



TECHNOLOGIES USED

- PYTHON
 - PANDAS
 - NUMPY
 - MATPLOTLIB
 - SEABORN
 - SCIKIT-LEARN
 - STREAMLIT
 - GITHUB
- 



DATASET DETAILS

- INPUT FEATURES:
 - STUDY HOURS
 - ATTENDANCE
 - PREVIOUS MARKS
 - OUTPUT:
 - FINAL MARKS PREDICTION
- 

METHODOLOGY

DATA COLLECTION



DATA PREPROCESSING



EXPLORATORY DATA ANALYSIS



MODEL TRAINING



MODEL EVALUATION



PREDICTION

MACHINE LEARNING MODEL

ALGORITHM USED:

➤ **LINEAR REGRESSION**

➤ REASON:

LINEAR REGRESSION IS USED TO PREDICT CONTINUOUS VALUES LIKE STUDENT MARKS.

RESULTS

THE MODEL SUCCESSFULLY PREDICTS FINAL MARKS BASED ON INPUT STUDENT INFORMATION.

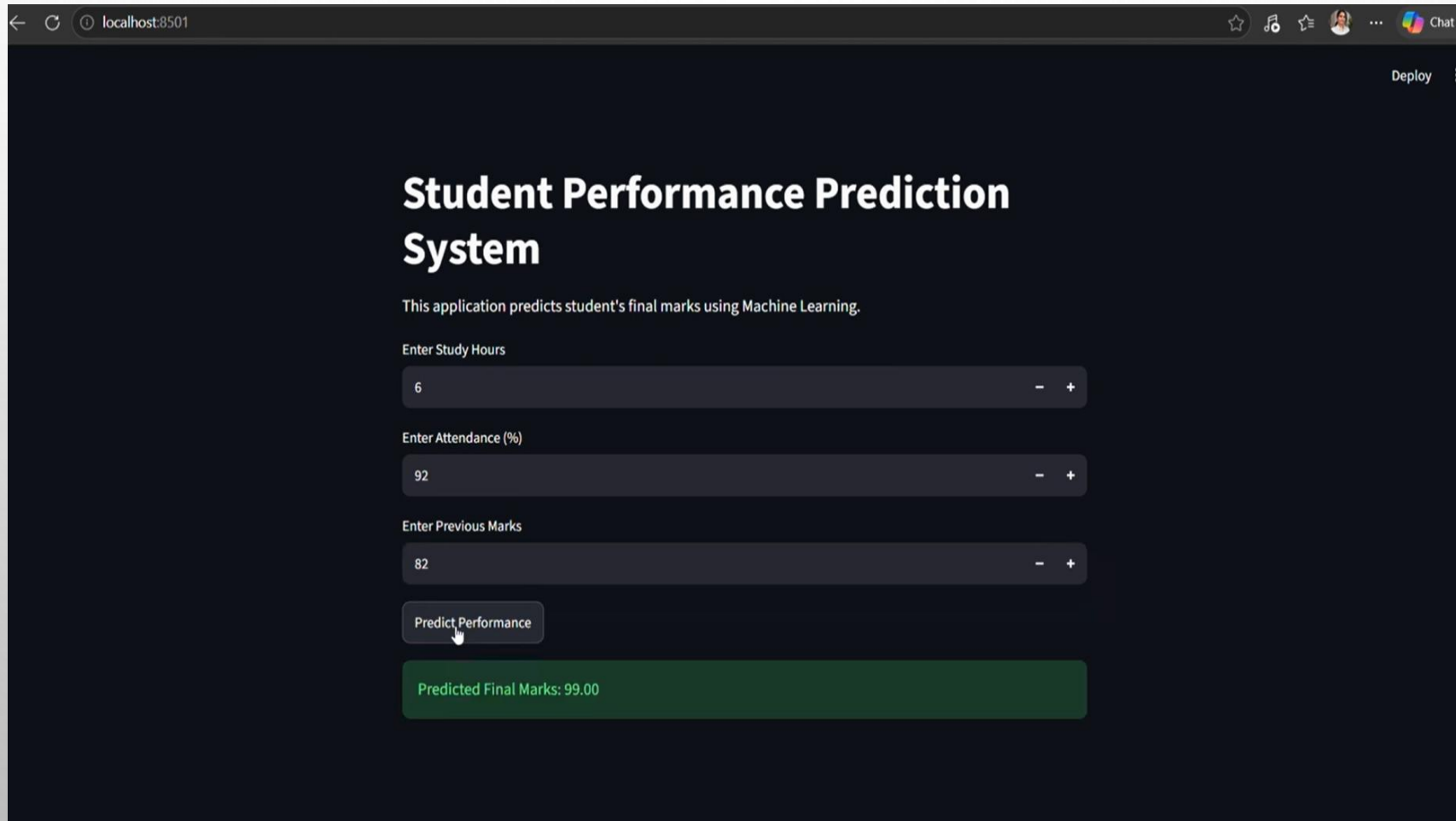
➤ EVALUATION METRICS:

➤ MAE

➤ MSE

➤ R2 SCORE

APPLICATION INTERFACE



The screenshot shows a web browser window with the address bar displaying 'localhost:8501'. The page has a dark theme and features a title 'Student Performance Prediction System' in white. Below the title is a subtitle: 'This application predicts student's final marks using Machine Learning.' The form contains three input fields with labels 'Enter Study Hours', 'Enter Attendance (%)', and 'Enter Previous Marks'. Each field has a numeric value (6, 92, and 82 respectively) and a minus/plus control. A 'Predict Performance' button is located below the third field. At the bottom, a green box displays the result: 'Predicted Final Marks: 99.00'. A 'Deploy' button is visible in the top right corner of the application area.

localhost:8501

Deploy

Student Performance Prediction System

This application predicts student's final marks using Machine Learning.

Enter Study Hours

6 - +

Enter Attendance (%)

92 - +

Enter Previous Marks

82 - +

Predict Performance

Predicted Final Marks: 99.00

CONCLUSION

- THE STUDENT PERFORMANCE PREDICTION SYSTEM DEMONSTRATES HOW MACHINE LEARNING CAN BE USED TO ANALYZE ACADEMIC DATA AND PREDICT STUDENT OUTCOMES.